

Integrated M.C.A (SEM-III)

Examination – Dec.-2023

7432

IMCA-302

System Programming

Time Allowed: Three Hours

Max. Marks: 70

Min. Pass Marks: 28

Note: This question paper contains three sections as under:

Section - A

Max. Marks - 10

This section shall have 01 compulsory question comprising 10 questions (maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

Section - B

Max. Marks - 25

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section - C

Max. Marks - 35

This section will carry 35 marks with five long answer type questions comprising one compulsory question (i.e. question no. 07) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions including one compulsory question (answer about in 400 words) out of remaining four questions.

Section-A

Q.1

- (i) List any two assembler directives.
- (ii) List any two advantages of assembly language.
- (iii) What is Declaration Statement? Give example.
- (iv) Differentiate between macro and subroutine.
- (v) Define Macro definition table.
- (vi) Write the basic tasks that can be performed by macro-instruction processor.
- (vii) What are the functions of the loader?
- (viii) Define Relocation factor.
- (ix) What is intermediate form?
- (x) What is a token? Give example.

Section-B

Q.2 Explain the general machine structures with neat diagram.

OR

Explain open subroutine and close subroutine with suitable example.

Q.3 Depict diagrammatically how a language is processed?.

OR

Explain macro instructions defining macros.

Q.4 Explain compile-and-go loader with a neat diagram.

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(Contd....

OR

List the phases that constitute the front end of a compiler?

Q.5 Explain different instruction formats of IBM360/370 machine.

OR

Explain and discuss one pass micro processor.

Q.6 What are the applications of structured programming to construct of system software tools? List each of them.

OR

Explain Functional modularity with suitable example.

Section-C

Q.7 What are the functions of analysis and synthesis phases of the compiler?

Q.8 Define binder. What are the classes of binders? Explain.

Q.9 What do you mean by Static Linking? What do you mean by dynamic loading? Explain.

Q.10 Write down all the steps of Pass 1 and Pass 2 of Linking? Explain.

Q.11 Which amongst the one pass and two assembler is better and why? Explain.

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